Zhenxi Gao

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EDUCATION

Nanjing University 09/2025-09/2027

Language Intelligence Processing Laboratory (LipLab), Master's Degree in Software Engineering, GPA 4.0/5.0

Nanjing University of Aeronautics and Astronautics 09/2021-06/2025

College of Computer Science and Technology, Bachelor's Degree in Internet of Things Engineering, GPA 3.9/5.0, Rank 6/63

PUBLICATION

Yang Y., Li C., Jin J., **Gao Z.**, Lin H., Zhang L., & Luo B. (2026). WANDER: Bridging Human Workflow and Automated Computer Use via Naturalistic Demonstrations and Enhanced Retrieval. *AAAI 2027*. Conference submission. (Link)

Gao Z. (2025). LH-Net: A Lightweight Heterogeneous Convolutional Network for RSSC. *WCI3DT 2024. Smart Innovation, Systems and Technologies,* vol 418. Springer. (Link)

ACADEMIC PROJECTS

MURAL-Presenter: Multi-agent Unified Revision-Aware Authoring for Long-horizon Presentations 06/2026-Present

ICLR (CCF-A) Research Project | SenseTime

Developed a skill-driven multi-agent framework for full-lifecycle presentation authoring that addresses cross-stage dependencies, cross-slide consistency, and iterative revision through shared deck-level state and impact-aware coordination.

- Contributed to the development of MURAL-Presenter, a skill-driven multi-agent framework for full-lifecycle presentation authoring, addressing cross-stage dependencies, cross-slide consistency, and iterative revision via shared deck-level state and impact-aware coordination
- Led THREAD-Bench, a benchmark that evaluates long-horizon capability through dependency preservation across multi-step, multi-tool, and multi-agent workflows, assessing requirement retention, factual accuracy, and visual consistency at Knowledge, Deck, and Page levels
- Trained MURAL-Lite by filtering high-quality decision-level revisions from teacher-model trajectories, enabling a lightweight model to acquire generation, error correction, and revision capabilities through process supervision
- Conducted systematic evaluations and key component ablations across proprietary and open-source models, validating the proposed framework's effectiveness in long-range consistency, error recovery, and multi-round revision

WANDER: Bridging Human Workflow and Automated Computer Use via Naturalistic Demonstrations and Enhanced Retrieval 03/2026-07/2026

AAAI 2027 (CCF-A) Research Project (GitHub)

Proposed a human-trajectory retrieval-augmented computer-use agent (CUA) that converts historical human operation trajectories into reusable structured knowledge, so as to mitigate the common planning drift and unstable operation granularity in general GUI agents during long-horizon desktop tasks.

- Designed and implemented a multi-module execution pipeline (Reflection-Planner-Grounding-Code Agent) that forms a closed loop from screenshot perception to task planning, action generation, UI positioning, and PyAutoGUI execution, and reduced error accumulation in long-horizon GUI tasks through a reflection-based control mechanism
- Built a canonical unit (CU) knowledge base, abstracting raw human trajectory steps into parameterized, reusable units; stored the multi-path statistics and historical transition relationships between CUs in a transition table
- Proposed a dual-channel retrieval-augmented planning method that combines FAISS on the current interface state with high-frequency successor operation recall from the transition table, improving experience reuse and planning stability for multi-step GUI tasks

SE-Explorer Agent: A Lightweight ReAct Agent for Software Engineering Tasks 11/2025-02/2026

Huawei Collaborative Research Project (GitHub)

Reconstructed the traditional RAG system that has fixed workflows and lacks interaction with code repositories into a lightweight ReAct-style agent loop, enabling the agent to dynamically perform knowledge retrieval, code localization, evidence collection, solution generation, and verification based on task states.

- Enhanced the search_docs tool via RAG-Anything to parse and retrieve body text, tables, formulas, and figures from software engineering papers in PDF/HTML formats; integrated Hybrid RAG's document retrieval, code search, file reading, shell/grep operations, patch generation, and test suggestion into composable tools for dynamic agent calling
- Designed trajectory logging, evidence memory, and verifier mechanisms to record thought-action-observation execution traces, structurally store multi-round tool calling results, and perform evidence consistency checks, enhancing execution

traceability and suppressing unfounded generation

- Constructed a small SE evaluation dataset based on SWE-QA-Pro and designed three types of demos: document-augmented question answering, code localization, and lightweight fixing; compared Direct LLM, Naive RAG, Hybrid RAG, and Agent modes, demonstrating the agent's advantages in complex cross-document/cross-file software engineering tasks

GenJudge: An Explainable Multi-Model Multimodal AIGC Detection Framework 03/2025-09/2025

Project supported by the National Natural Science Foundation of China (GitHub)

Designed a multi-model framework that simulates expert review mechanisms for detecting AI-generated content (AIGC) in image-text pairs, addressing the poor cross-domain generalization of existing single-model detection approaches.

- Built a multimodal evidence pool integrating image forensics (e.g. noise residuals), text analysis (e.g. perplexity), and cross-modal logic (physical/temporal logic)
- Designed a two-stage review mechanism: Stage 1 performs distributed judgment via Qwen2.5-VL and InternVL models; Stage 2 introduces a "critique and revision" process that optimizes analysis weights for disputed evidence through inter-model interaction, producing interpretable judgment labels and evidence chains via fuzzy logic aggregation
- Constructed a high-quality multi-domain dataset comprising 20,000 samples, covering general, news, satellite, and document domains and four fine-grained categories (HH/AH/HA/AA)
- Achieved an average accuracy of 0.632 in cross-generator and cross-domain testing, significantly outperforming baseline methods such as HAMMER and InternVL, and effectively addressing the performance drops of single models like Qwen in specific domains (e.g. news)

INTERNSHIPS

SenseTime, Multimodal LLM Department, Algorithm Intern 06/2026-Present

- Developed a query synthesis pipeline for the Static HTML PPT Agent, constructing a reusable training/evaluation sample pool from real backflow data with PII masking, usability filtering, and multi-dimensional metadata modeling; ensured query naturalness via LLM rewriting and rule-based validation
- Designed coverage-controlled query generation to balance domain/style/complexity distributions, with LLM review and rule-based validation for consistency, schema compliance, and anti-leakage

GoLaxy, R&D Center, Algorithm Intern 05/2025-08/2025

- Built a Qwen2.5-based pipeline for 600K financial news records, performing entity alignment, long-document summarization, and information extraction; fine-tuned a UIE-based model for company-business description relation extraction
- Developed a multimodal speech processing system integrating speaker diarization, voiceprint recognition, and speech transcription, and deployed it on internal video datasets

OTHER INFORMATION

- Programming Skills: Python, C/C++, Java, Linux, Docker, Git
- Language Proficiency: Chinese (mother tongue), English (CET-6 591)
- Honors: Honorable Mention in the 2024 Mathematical Contest in Modeling, Third Prize in the C/C++ Programming University Group A of the 14th Lanqiao Cup National Software and Information Technology Professional Talent Competition (Jiangsu Division) in 2023, Second Prize in the 14th Chinese Mathematics Competition for College Students (Non-Mathematics Category) in 2023